

No. of Functional Features Included in the Solution

Date	15 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-being
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form	Allows users to enter Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and GNI per Capita.	Frontend (HTML/CSS)	Done	High
2	Data Validation	Validates user input before sending it for prediction.	Flask Backend	Done	Medium
3	HDI Prediction Engine	Predicts the Human Development Index using the trained Linear Regression model.	Machine Learning Model	Done	High
4	Development Category Classification	Classifies the predicted HDI as Low, Medium, High, or Very High.	Prediction Logic	Done	High
5	Results Display	Displays the HDI score and development category in a responsive results page.	Flask Templates	Done	High
6	Model Loading	Loads the trained .pkl model during application startup for predictions.	Backend	Done	Medium
7	Responsive User Interface	Provides a clean and responsive interface that works across different screen sizes.	HTML/CSS	Done	Medium
8	GitHub Documentation	Includes README, screenshots, setup instructions, and source code for easy project usage.	Documentation	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6 (Input Form, Data Validation, HDI Prediction, Category Classification, Results Display, Model Loading)
Additional / Nice-to-Have Features	2 (Responsive UI, GitHub Documentation)
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Home Page, Input Form, Result Page
2	Backend / Logic	4	Data Validation, Model Loading, Prediction Engine, HDI Classification
3	Database / Storage	1	CSV Dataset and Trained .pkl Model
4	API / Integration	1	Flask Routes for Prediction Requests
5	Security / Authentication	0	Not implemented (authentication is outside the current project scope)